

Behavior Landscape Guided LLM For Policy Search

Prompted Policy Search:

Use LLM as optimizer to find optimal policy θ^* which minimize $J(\theta) = E[r]$. By providing task description and search history at each iteration $H_t = (\theta_i, J(\theta_i)) : 1 \leq i \leq t$, LLM will decide which policy to search next. We can consider this as sequential decision making.

Our Proposal:

Existing Gap: Existing RL (include Gradient Policy method and ProPS) relies solely on reward signals, which may be uninformative in sparse-reward environments (like Mountain Car...)

We can use VLM as the trajectory analyzer. By passing into VLM frames of a trajectory, VLM will feedback by the behavior description of the policy. Furthermore, we just not only use VLM only 1 policy but also its neighbor, by adding to its dimension random small number. Totally, the iteration looklike, we have 3 anchor: best, worst and current policy. We will describe the behavior of policy and its neighbor by using VLM, then pass them into LLM to optimize.

Full Pseudocode:

Training Loop

for $t = 1, \dots, N_{episodes}$

$\{\theta_{cur}, \theta_{best}, \theta_{worst}\}$

are selected as anchors.

For each anchor θ_a :

$\tau_a, F_a = \text{Rollout}(E, \theta_a)$

$analysis_a = \text{VLM}(F_a, D, \theta_a)$

Sample neighbors

$\theta_j = \theta_a + \Delta_j$

and analyze them with VLM.

Construct neighborhood summary

Ψ_t

LLM proposes next policy

$\theta_{t+1} = \text{LLM}(\Gamma, P, \Psi_t, D)$

Evaluate policy

$r_{t+1} = \text{Evaluate}(E, \theta_{t+1})$

PROPSV_geminiVLM

Problem							
hopper							
mountaincar	-111.200	-114.2	-109.3	-105.55	-115.65	-113.7	-162.5
mountaincarcontinuous	99.239	99.23	99.14	98.93	99.17	99.17	98.9
nav	2551.725						
pong	2.850						
reacher	-7.401	-7.47	-7.51	-6.81	-7.33	-6.71	-6.72
swimmer	307.756	273.5	299.63	353.56	294.26	347.79	344.3
frozenlake	1.00	0.95	0.95	0.95	0.95	0.95	0.95

Problem							
cliffwalking	-13.00	-64.35	-53.3	-55.05	-63.0	-60.45	-65.7!
acrobot	-74.400	-77.95	-70.9	-79.7	-73.75	-71.85	-74.8
lunarlander	268.652	-31.71	274.85	219.01	-105.73	156.41	6.45
inverted pendulum							
double inverted pendulum							
walker							

PROPSV_GemmaVLM

Problem							
hopper							
mountaincar	109.65	113.75	108.65	108.2	117.85	111.3	111.7
mountaincarcontinuous	99.19	98.87	99.05	99.21	99.28	99.32	99.15
nav	2714.5	2881.95	4243.475	346.82	2482.85	1482.53	3161.0
pong							
reacher	-5.99	-7.99	-7.9	-6.74	-8.17	-7.47	-7.58
swimmer	343.17	348.59	345.97	346.53	300.2	351.96	235.5
frozenlake	1.00	1.00	1.00	0.95	0.95	0.95	0.95
cliffwalking (100e)	-64.75	-61.9	-60.65	-67.2	-63.0	-67.15	-68.3
acrobot							
lunarlander	-123.61	276.03	-107.05	286.41	-69.69	289.52	34.11
inverted pendulum	1000.0	1000.0	1000.0	1000.0	1000.0	1000.0	1000.0
double inverted pendulum							
walker							

PROPSV_w/o_history_gemmaVLM

Problem							
hopper							
mountaincar							
mountaincarcontinuous							
nav							
pong							
reacher	-8.44	-7.26	-8.79	-7.77	-5.85	-7.57	-8.28
swimmer							
frozenlake							
cliffwalking							
acrobot							

Problem							
lunarlander							
inverted pendulum							
double inverted pendulum							
walker							

OPEN AI ES

Problem							
hopper	962.243	1039.606	962.243	1045.055	1049.767	1069.425	1066.1
mountaincar							
mountaincarcontinuous							
nav	-147.500000	-145.500000	-193.450000	-101.375000	2397.900000	-193.975000	-228.1
pong	2.850	1.250	1.100	0.750	0.550	0.650	1.050
reacher							
swimmer							
frozenlake							
cliffwalking							
acrobot							
lunarlander							
inverted pendulum	87.891	91.649	157.268	90.653	129.575	88.333	4.006
double inverted pendulum	91.208	91.287	91.682	91.630	99.149	99.981	88.50
walker	-13.087456	-0.566577	-12.910347	-6.995744	-1.706098	-4.942753	-4.02

PROPS

Problem					
hopper					
mountaincar	-182.900	-114.5	-114.4	-108.25	-110.9
mountaincarcontinuous	94.168	98.96814244335538	95.10910452934351	99.15495856046735	99.0220904562
nav	392.325	-31.55	2563.15	2758.1	2514.275
pong	3.000				
reacher	-7.298	-7.620853845951426	-7.384176842875911	-7.52757733074296	-8.14995354987
swimmer	161.465	219.68589240287082	162.1556306436805	265.72806230538606	252.5414295521
frozenlake	0.9	0.9	0.9	0.9	0.9
cliffwalking	-62.800	-60.55	-81.9	-63.75	-71.85
acrobot	-77.850	-74.75	-78.55	-73.4	-86.0
lunarlander	-36.069	-75.24757897957662	-40.94392627629045	-50.429888406322874	261.61111912246
inverted pendulum	1000.0	1000.0	204.65	1000.0	808.6
double inverted pendulum	114.68348301887697	109.17171683197543	169.50703821213537	147.01886594751724	342.338626460
walker					

CMA ES

Problem							
hopper	1097.638203	2578.138752	1766.232909	1119.890677	1162.300574	1883.087445	2301.
mountaincar	-200.000000	-200.000000	-200.000000	-200.000000	-200.000000	-200.000000	-200.
mountaincarcontinuous	98.794665	-0.000045	99.337971	-0.002120	-0.001991	98.728224	12.16
nav	-203.375000	121.575000	-215.550000	-111.45000	3348.925000	1.325000	-229.
pong	0.500000	0.600000	0.500000	0.650000	3.000000	0.600000	0.700
reacher	-8.993837	-11.341409	-8.753249	-9.259011	-11.619965	-9.238485	-8.66
swimmer	142.486439	-5.246859	-7.802748	357.836918	357.650224	: 54.069142	359.3
frozenlake	0.150000	0.200000	0.000000	0.150000	0.150000	: 0.250000	0.250
cliffwalking	-189.550000	-189.550000	-100.000000	-975.300000	-169.300000	-100.000000	-632.
acrobot	-122.750000	-72.200000	-500.000000	-71.600000	-68.800000	-500.000000	-112.
lunarlander							
inverted pendulum	1000.000000	2.000000	30.900000	1000.000000	5.800000	1000.000000	2.000
double inverted pendulum	87.846052	137.775113	178.278985	9359.843703	152.447929	171.055866	71.57
walker	-31.223300	-4.101530	183.163140	-7.942113	-12.130742	-9.493010	-10.1

ARS

Problem							
hopper	268.517	316.532	956.414	4.69	125.848	6.875	900.1
mountaincar	-200	-200	-200	-200	-200	-200	-200
mountaincarcontinuous	-0.005	-0.006	-0.004	98.979	-0.132	-0.029	-2.66
nav	-171.700000	-41.600000	-180.150000	-161.325000	-159.900000	-172.700000	1.1750
pong	0.350	0.200	0.100	0.250	0.100	0.250	0.200
reacher	-8.978076	-9.097515	-12.414031	-13.479	-11.196	-35.638	-40.2
swimmer	51.820	36.059	33.751	307.808	-5.239	37.507651	-18.4
frozenlake							
cliffwalking							
acrobot	-500.000000	-167.550000	-500.000000	-500.000000	-500.000000	-500.000000	-500.
lunarlander							
inverted pendulum	115.800000	2.000000	4.000000	2.000000	1000.000000	36.150000	2.000
double inverted pendulum	89.269722	23.579702	23.587026	23.997287	44.715535	61.277883	792.3
walker	-46.437514	: -20.110186	-36.290657	-47.287905	-24.958954	-54.331889	-19.2

(mu, lambda) ES

Problem							
hopper	1111.665245	1164.535718	1027.597557	1096.289609	1026.887806	2432.252473	1084.
mountaincar	-200	-200	-200	-200	-200	-200	-200
mountaincarcontinuous	-0.008	84.131	-0.096	-0.007	-0.050	-0.005	99.17
nav	1985.975000	-248.925000	-238.425000	-214.025000	-99.875000	2801.450000	-164.0
pong	0.200	0.200	0.200	1.700	0.200	0.550000	0.550

Problem							
reacher	-10.827935	-11.627046	-116.297144	-10.225912	-9.698219	-10.828273	-14.4
swimmer	120.139576	34.354194	53.006763	356.763594	119.367674	138.926481	352.5
frozenlake							
cliffwalking							
acrobot	-83.250	-86.900	-482.350	-500.000	-158.050	-159.000	-86.5
lunarlander							
inverted pendulum	236.450000	2.000000	1000	1000	1000	402.900000	64.7
double inverted pendulum	27.124842	23.557055	27.332174	281.760126	82.629333	101.388268	49.38
walker	-8.622864	-22.084030	-14.488109	-4.420381	-11.643680	-7.485784	-6.54

All result

Problem	PROPSV_geminiVLM	PROPSV_GemmaVLM	PROPSV_w/o_history_gemmaVLM	PROPSV_w/o_env_description
hopper		675.29 ± 364.17		
mountaincar	-117.06 ± 16.25	-112.01 ± 2.85	-133.95 ± 28.20	-142.055 ± 40.751
mountaincarcontinuous	99.07 ± 0.15	99.16 ± 0.12	99.14 ± 0.08	
nav		2284.0575 ± 1252.68		
pong		2.96 ± 0.10		
reacher	-7.28 ± 0.38	-7.358 ± 0.664	-7.68 ± 0.81	
swimmer	320.74 ± 27.49	309.36 ± 51.83		206.325 ± 103.107
frozenlake	0.965 ± 0.023	0.965 ± 0.023	0.9050 ± 0.0497	
cliffwalking	-58.31 ± 16.21	-66.86 ± 8.47		
acrobot	-75.42 ± 2.72	-75.42 ± 2.72	-79.30 ± 10.65	
lunarlander	73.24 ± 145.46	104.83 ± 168.86		
inverted pendulum		1000.00 +- 0		
double inverted pendulum		982.31 +- 2482.35		
walker		477.78 ± 198.11		